

Quadratic functions and technology



- 1** Graph each function using a calculator or other technology then answer the following questions:
- i** State the coordinates of the vertex.
 - ii** State the coordinates of the y -intercept.
- a** $y = 4x^2 - 64x + 263$ **b** $y = 4x^2 + 48x + 135$
c $y = -4x^2 - 48x - 140$ **d** $y = -3x^2 + 12x - 13$
e $y = 3x^2 + 3x + 9$ **f** $y = 4x^2 + 3x - 1$
- 2** Graph each function using a calculator or other technology then answer the following questions:
- i** State the coordinates of the vertex of the graph.
 - ii** Are there any x -intercepts? If yes, find the x -value(s) of the x -intercept(s).
 - iii** For $x > 0$, is the parabola increasing or decreasing?
- a** $y = 4x^2 + 16$ **b** $y = -3x^2 - 12$ **c** $y = -5x^2 + 125$ **d** $y = 4x^2 - 64$
- 3** State whether the following parabolas have x -intercepts:
- a** $y = (x - 7)^2 + 4$ **b** $y = (x - 7)^2 - 4$
c $y = -(x - 7)^2 + 4$ **d** $y = -(x - 7)^2 - 4$
- 4** Find the value of c if the parabola $y = x^2 + 10x + c$ has exactly one x -intercept.
- 5** Graph each function using a graphing calculator then answer the following questions:
- i** Find the x -value(s) of the x -intercept(s).
 - ii** Find the y -value of the y -intercept.
 - iii** State the coordinates of the vertex.
- a** $y = \left(x + \frac{5}{2}\right)^2$ **b** $y = -(x - 3)^2 + 4$
c $y = 9 - (x - 4)^2$



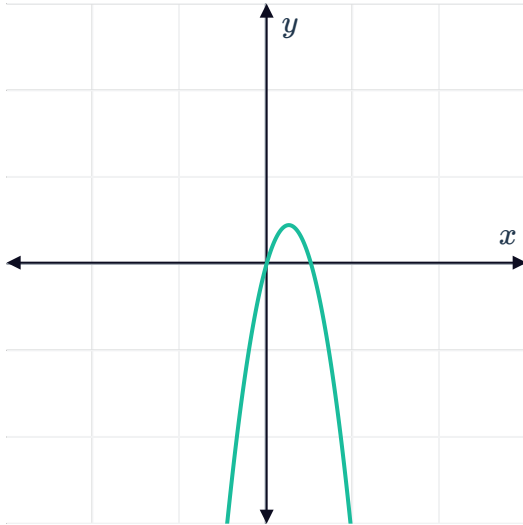
- 6 Consider the parabolas with equations $f(x) = x^2 + 2x + 4$ and $g(x) = x^2 - 2x + 4$:
- Using a graphing calculator, state the coordinates of the vertex of each parabola:
 - For $f(x)$
 - For $g(x)$
 - How far apart are the vertices of the two parabolas?
- 7 Will the curve $y = -x^2 - 5x + 1$ reach a maximum or a minimum value as it turns?
- 8 Graph $y = x^2 - 4$ using a calculator or other technology then answer the following questions:
- Does it have a minimum or maximum y -value?
 - State the coordinates of the vertex.
- 9 Using a graphing calculator or other technology, find the minimum value of y for the quadratic function $y = x^2 + x - 20$.
- 10 Graph $y = x^2 + 6.2x - 7$ using a calculator or any technology then answer the following questions:
- State the equation of the axis of symmetry.
 - Find the minimum value of y .
- 11 Graph the following functions using a graphing calculator then answer the following questions:
- Find the x -value(s) of the x -intercept(s).
 - Find the y -value of the y -intercept.
 - State the equation of the axis of symmetry.
 - State the coordinates of the vertex.
- $y = x(6 - x)$
 - $y = x^2 + 2x$
 - $y = -x^2 - 4x$
 - $y = (x + 1)(x - 3)$
 - $y = x^2 + 2x - 3$
 - $y = -2x^2 + 16x - 24$

12 Consider the function $y = -0.68x^2 + 3.48x$:



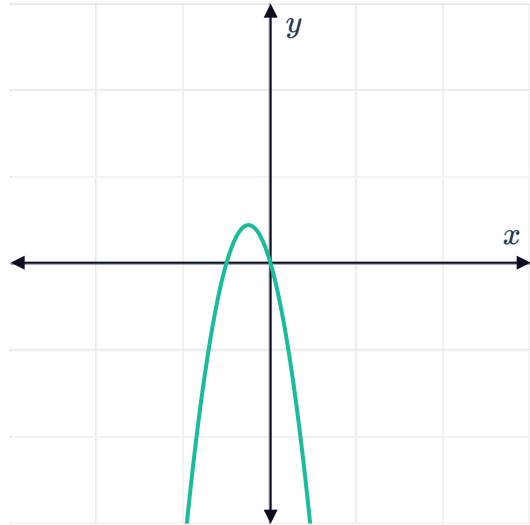
- a Find the x -coordinate of the vertex, correct to two decimal places.
- b Find the y -coordinate of the vertex, correct to two decimal places.
- c State whether the following are possible viewing window on your calculator that shows the vertex and the x -intercepts:

i



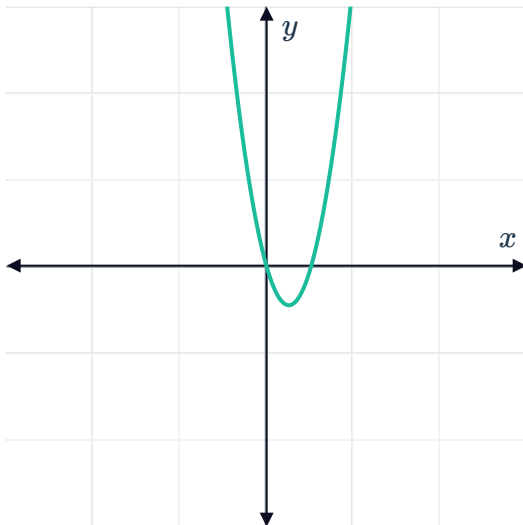
–20, 30, 5 by –20, 30, 5

ii



–20, 30, 5 by –20, 30, 5

iii



–20, 30, 5 by –20, 30, 5

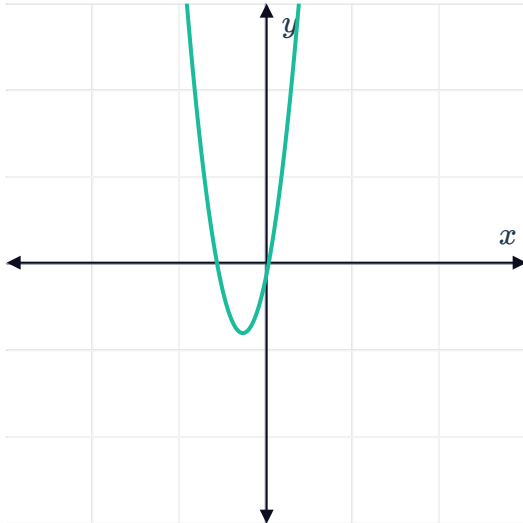
- d Use a graphing calculator to find the x -intercepts, correct to two decimal places.

13 Consider the function $y = 0.91x^2 - 5x - \sqrt{5}$



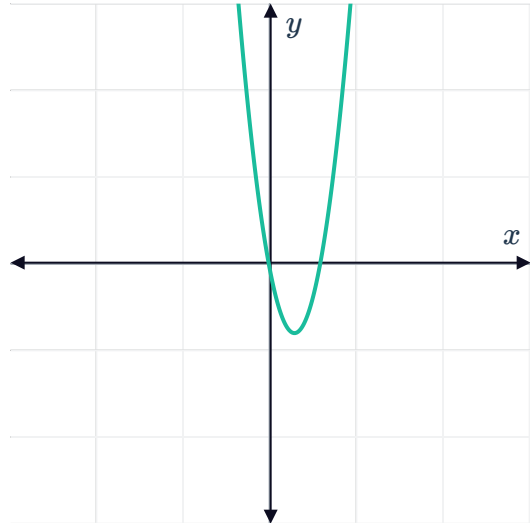
- a Find the x -coordinate of the vertex, correct to two decimal places.
- b Find the y -coordinate of the vertex, correct to two decimal places.
- c State whether the following are possible viewing window on your calculator that shows the vertex and the x -intercepts:

i



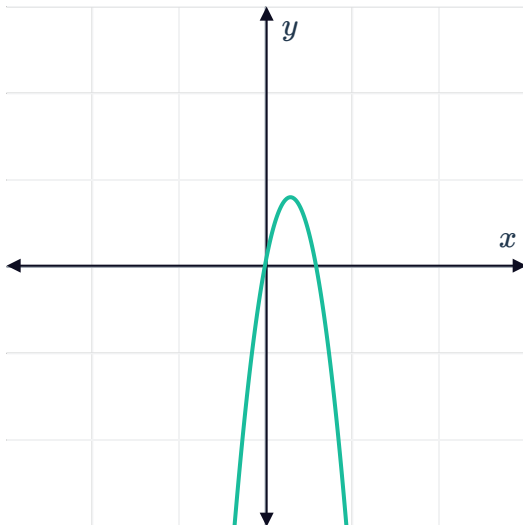
$-20, 30, 5$ by $-30, 20, 5$

ii



$-20, 30, 5$ by $-30, 20, 5$

iii



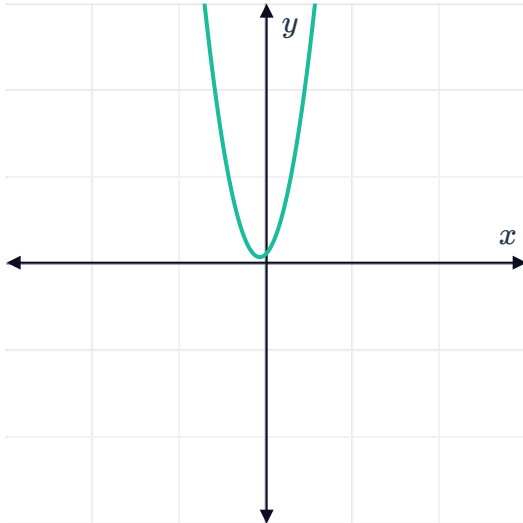
$-20, 30, 5$ by $-30, 20, 5$

- d Use a graphing calculator to find the x -intercepts to two decimal places.

14 Consider the function $y = -0.72x^2 + \sqrt{5}x + 4$ 

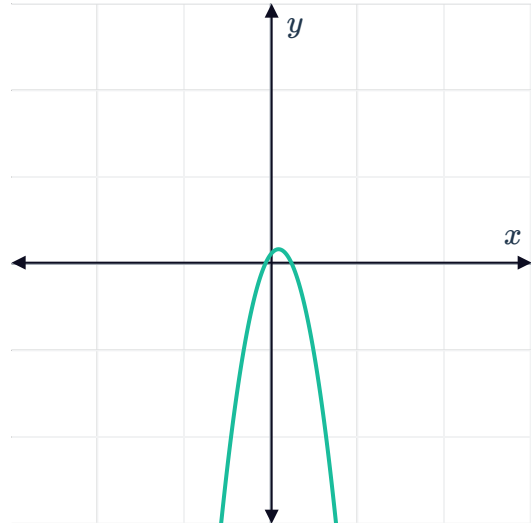
- a Find the x -coordinate of the vertex, correct to two decimal places.
- b Find the y -coordinate of the vertex, correct to two decimal places.
- c State whether the following are possible viewing window on your calculator that shows the vertex and the x -intercepts:

i



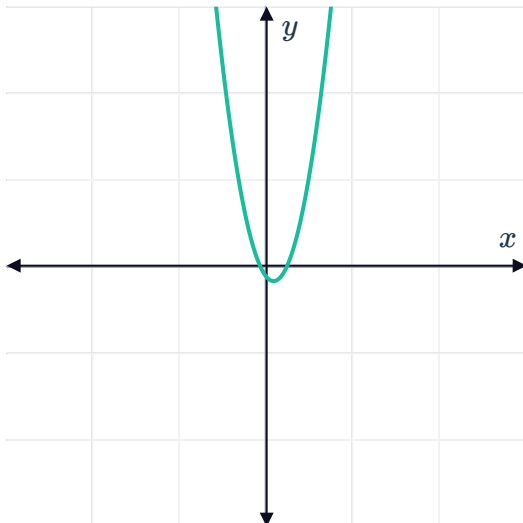
–20, 30, 5 by –20, 30, 5

ii



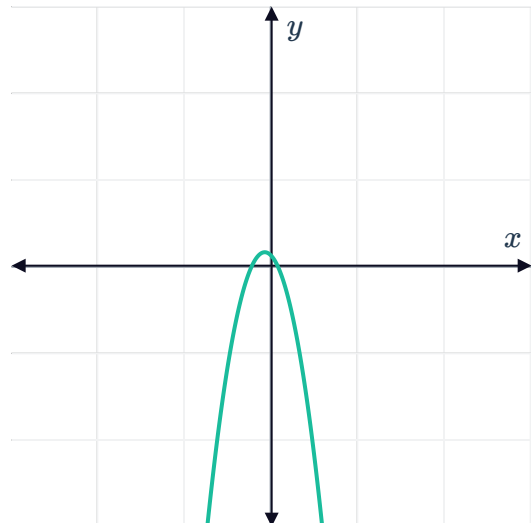
–20, 30, 5 by 0, 30, 5

iii



–20, 30, 5 by –20, 30, 5

iv



–20, 30, 5 by –20, 30, 5

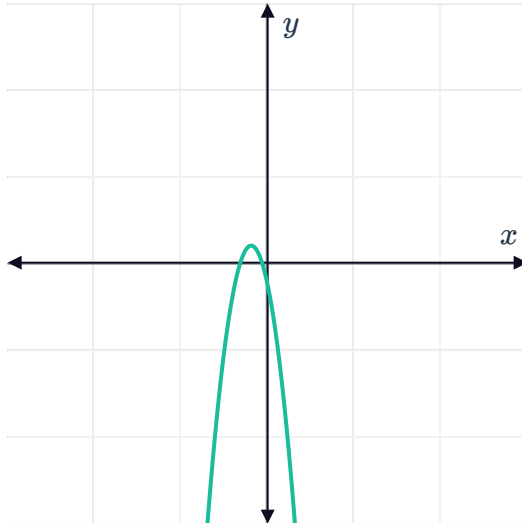
- d Use a graphing calculator to find the x -intercepts, correct to two decimal places.

15 Consider the function $y = -\sqrt{3}x^2 - 4.79x - 2.4$



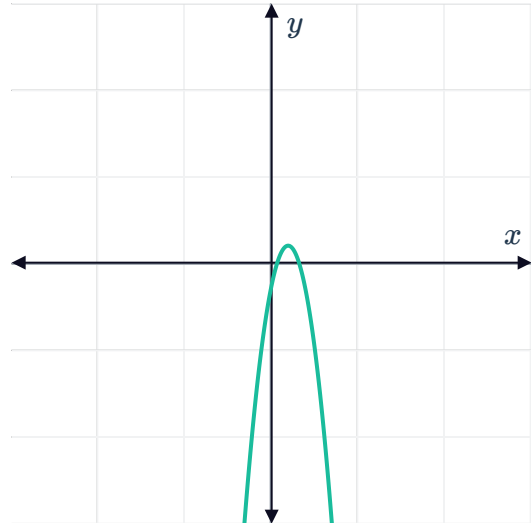
- a Find the x -coordinate of the vertex, correct to two decimal places.
- b Find the y -coordinate of the vertex, correct to two decimal places.
- c State whether the following are possible viewing window on your calculator that shows the vertex and the x -intercepts:

i



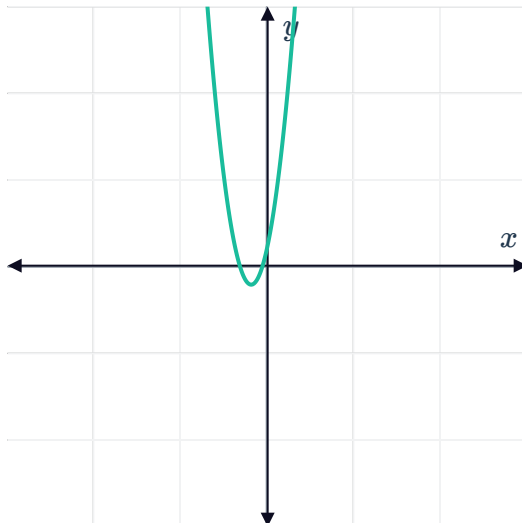
–30, 20, 5 by –20, 30, 5

ii



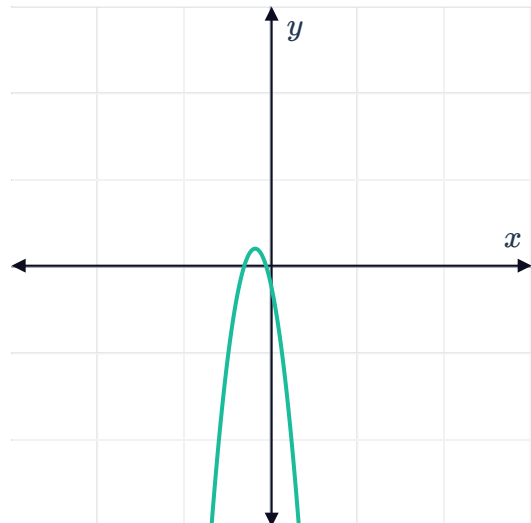
–30, 20, 5 by –20, 30, 5

iii



–30, 20, 5 by –20, 30, 5

iv



–30, 20, 5 by 0, 30, 5

- d Use a graphing calculator to find the x -intercepts, correct to two decimal places.

16 How do we shift the graph of $y = x^2$ to get the graph of $y = x^2 - 5$?

17 How do we shift the graph of $y = x^2$ to get the graph of $y = (x - 3)^2$?



- 18** Using a calculator or other technology, graph functions $y = x^2$, $y = 4x^2$, and $y = 6x^2$. For $y = ax^2$, as a increases, how does it change the graph of $y = x^2$?
- 19** Using a graphing calculator, graph $y = x^2$ and write the equation needed for the following translations:
- a** $y = x^2$ shifted 2 units up
 - b** $y = x^2$ reflected about the x -axis
 - c** $y = x^2$ shifted 2 units up then reflected about the x -axis
 - d** $y = x^2$ is translated 2 units to the left and 2 units up
- 20** A parabola of the form $y = (x - h)^2 + k$ is symmetrical about the line $x = 2$, and its vertex lies 3 units above the x -axis. Write the equation needed to graph the parabola into a graphing calculator.
- 21** Solve each of the following equations by splitting the equation into two functions and graphing them using a calculator or other technology:
- a** $7x^2 = 8x$
 - b** $2x \left(x - \frac{5}{2} \right) = 3$
 - c** $2x(x - 1) = -x + 3$
 - d** $\frac{5 + 19x}{4x} = x$
 - e** $(x + 1)(x - 2) = -3x - 3$
 - f** $\frac{x}{x + 2} + \frac{9}{x - 2} = \frac{6}{x^2 - 4}$
- 22** Graph each function using a calculator or other technology then determine the number of unique solutions to the corresponding equation:
- a** Function: $f(x) = (x - 9)^2$
Equation: $(x - 9)^2 = 0$
 - b** Function: $f(x) = (x - 3)(x - 2)$
Equation: $(x - 3)(x - 2) = 0$
 - c** Function: $f(x) = -x^2 + 8x - 17$
Equation: $-x^2 + 8x - 17 = 0$
- 23** Find the solution(s) of $3x^2 - 11x + 10 = 0$, correct to three decimal places.
- 24** Find the zero(s) of $f(x) = 3.83x^2 - 2.78x - 5.21$, correct to three decimal places.



Additional questions

- 25** The Millau Viaduct in Millau, Southern France, is 270 m tall. We can model the behaviour of objects falling from the viaduct using Galileo's formula for falling objects: $d = 4.9t^2$, where d is distance fallen in meters and t is time in seconds since the object was dropped. Graph this equation using a calculator or other technology then answer the following questions:
- a** State the coordinates of the vertex of the graph.
 - b** State what the answer in part (a) represent.
- 26** A rectangle is to be constructed with 60 m of wire. The rectangle will have an area of $A = 30x - x^2$, where x is the length of one side of the rectangle. Graph this equation using a calculator or other technology then answer the following questions:
- a** Find the x -values of the x -intercepts.
 - b** Find the x -value of the vertex.
 - c** State what the height of the parabola at $x = 15$ represent.
- 27** The kinetic energy E of a moving object is given by $E = \frac{1}{2}mv^2$, where m is its mass in kilograms and v is its speed in meters/second. Graph this equation for a vehicle with a mass of 1400 kg using a calculator or other technology. Then, calculate the velocity when the kinetic energy is 137 200 J.
- 28** An object launched from the ground has a height (in meters) after t seconds that is modelled by the equation $y = -4.9t^2 + 58.8t$. Graph this equation using a calculator or other technology then answer the following questions:
- a** Find the maximum height of the object.
 - b** After how many seconds is the object at its maximum height?
 - c** After how many seconds does the object return to the ground?



29 An object is released 900 m above ground and falls freely. The distance the object is from the ground is modelled by the formula $d = 900 - 4.9t^2$, where d is the distance in meters that the object falls and t is the time elapsed in seconds.

Graph this equation using a calculator or other technology then answer the following questions:

- a** Find d when $t = 0$.
- b** Explain what the answer in part (a) mean.
- c** Find the value of t when $d = 0$, correct to two decimal places.
- d** Explain what the answer in part (c) mean.

30 On the moon, the equation $d = 0.8t^2$ is used to find the distance an object has fallen after t seconds. On Earth, the equation is $d = 4.9t^2$.

- a** A rock is thrown from a height of 80 m on Earth. Graph the equation on a calculator or other technology to find the time taken to hit the ground to the nearest second.
- b** A rock is thrown from a height of 80 m on the Moon. Graph the equation on a calculator or other technology to find the time taken to hit the ground to the nearest second.